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play oxygenase role. Succulent and xerophytic plants show reverse stomatal rhythm as they open during night time and close during day time and follow CAM cycle. Night time produces malic acid and during day time malate is converted into pyruvate and produces CO_2 which is reduced to carbohydrates. Photosynthesis is affected by internal and external factors. Bacterial photosynthesis is the primitive type of photosynthesis and it involves only photosystem I.

CPZYDC

Evaluation

1. **Assertion (A):** Increase in Proton gradient inside lumen responsible for ATP synthesis

   Reason (R): Oxygen evolving complex of PS I located on thylakoid membrane facing Stroma, releases H+ ions

   a. Both Assertion and Reason are True.

   b. Assertion is True and Reason is False.

   c. Reason is True and Assertion is False.

   d. Both Assertion and Reason are False.

2. Which chlorophyll molecule does not have a phytol tail?

   a. Chl- a b. Chl-b c. Chl- c d. Chl -d

3. The correct sequence of flow of electrons in the light reaction is

   a. PS II, plastoquinone, cytochrome, PS I, ferredoxin.

   b. PS I, plastoquinone, cytochrome, PS II ferredoxin.

   c. PS II, ferredoxin, plastoquinone, cytochrome, PS I.

   d. PS I, plastoquinone, cytochrome, PS II, ferredoxin.

4. For every CO_2 molecule entering the C_3 cycle, the number of ATP & NADPH required

   a. 2ATP + 2NADPH

   b. 2ATP + 3NADPH

   c. 3ATP + 2NADPH

   d. 3ATP + 3NADPH

5. Identify true statement regarding light reaction of photosynthesis?

   a. Splitting of water molecule is associate with PS I.

   b. PS I and PS II involved in the formation of NDPH+H^+.

   c. The reaction center of PS I is Chlorophyll a with absorption peak at 680 nm.

   d. The reaction center of PS II is Chlorophyll a with absorption peak at 700 nm.

6. Two groups (A & B) of bean plants of similar size and same leaf area were placed in identical conditions. Group A was exposed to light of wavelength 400-450nm & Group B to light of wavelength of 500-550nm. Compare the photosynthetic rate of the 2 groups giving reasons.

7. A tree is believed to be releasing oxygen during night time. Do you believe the truthfulness of this statement? Justify your answer by giving reasons?

8. Grasses have an adaptive mechanism to compensate photorespiratory losses- Name and describe the mechanism.

9. In Botany class, teacher explains, Synthesis of one glucose requires 30 ATPs in C_4 plants and only 18 ATPs in C_3plants. The same teacher explains C_4 plants are more advantageous than C_3 plants. Can you identify the reason for this contradiction?

10. When there is plenty of light and higher concentration of O_2, what kind of pathway does the plant undergo?Analyse the reasons.